



MAT1320-Linear Algebra

Lecture Notes

Eigenvalues and Eigenvectors, Characteristic Polynomial,
Diagonalization, Cayley-Hamilton Theorem

Mehmet E. KÖROĞLU

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YILDIZ TECHNICAL UNIVERSITY, DEPARTMENT OF MATHEMATICS

mkoroglu@yildiz.edu.tr

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Eigenvalues and Eigenvectors

Eigenvalues and Eigenvectors

Definition

A scalar λ is called an **eigenvalue** of the $n \times n$ square matrix A if there is a nontrivial solution $\vec{x}$ of $A\vec{x} = \lambda\vec{x}$.

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Note: Note that an **eigenvector cannot be $\vec{0}$** , but an eigenvalue can be $0 \in \mathbb{R}$. If 0 is an eigenvalue of A , then there must be some nontrivial vector $\vec{x}$ for which $A\vec{x} = 0\vec{x} = \vec{0}$ which implies that A is not invertible.

$$A\vec{x} = \vec{0} \rightarrow A^{-1}A\vec{x} = A^{-1}\vec{0} \\ \boxed{\vec{x} = \vec{0}}$$

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The $n \times n$ square matrix A is invertible if and only if 0 is not an eigenvalue of A .

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Theorem

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$$V_\lambda = \{ \vec{x} : A\vec{x} = \lambda\vec{x} \}$$

Note: The eigenspace of the $n \times n$ matrix A corresponding to the eigenvalue λ of A is the set of all eigenvectors of A corresponding to λ .

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$$\begin{aligned} A\vec{x} &= \lambda\vec{x} \Rightarrow A\vec{x} - \lambda\vec{x} = \vec{0} \\ A\vec{x} - \lambda I_n \vec{x} &= \vec{0} \quad \Rightarrow (A - \lambda I_n) \cdot \vec{x} = \vec{0} \end{aligned}$$

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Definition

Let A be an $n \times n$ square matrix. Then the equation

$\det(A - \lambda I_n) = 0$ is called the characteristic equation of the matrix A and the result of the determinant $\det(A - \lambda I_n)$ is polynomial of the form $P_A(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_1\lambda + a_0$ is called characteristic polynomial of A .

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Note: $P_A(\lambda) = (-1)^n \lambda^n + (-1)^{n-1} \text{Tr}(A) \lambda^{n-1} + \dots + \det(A)$.

For example, for a 2×2 square matrix A

Mehmet E. KÖROĞLU

$$P_A(\lambda) = \lambda^2 - \text{Tr}(A)\lambda + \det(A)$$

Handwritten notes: $\lambda^2 - \lambda + 2$, $\text{Tr}(A)=1$, $\det(A)=-2$, $=0$.

Finding Eigenvalues and Eigenvectors

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1. Find characteristic polynomial $\det(A - \lambda I_n) = \underline{P_A(\lambda)}$ of A .

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Note 1: The eigenvalues of a triangular matrix are the entries on its main diagonal.

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{pmatrix} \rightarrow \underline{\text{Eigenvalues: } 1, 2, 3}$$

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Note 1: The eigenvalues of a triangular matrix are the entries on its main diagonal.

Note 2: If $P_A(\lambda)$ has multiple roots, then there exists multiple eigenvalues.

$$P_A(\lambda) = (x-1)^2(x+1) \quad \text{eigenvalues: } 1, -1,$$

Finding Eigenvalues and Eigenvectors

Example

Let $A = \begin{pmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{pmatrix}$. Find

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Finding Eigenvalues and Eigenvectors

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Finding Eigenvalues and Eigenvectors

$$\underline{A - \lambda \cdot \mathbb{I}_3} = \begin{pmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{pmatrix} - \lambda \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3-\lambda & 6 & -8 \\ 0 & -\lambda & 6 \\ 0 & 0 & 2-\lambda \end{pmatrix}$$

Example

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$$\rightarrow P_A(\lambda) = \begin{vmatrix} 3-\lambda & 6 & -8 \\ 0 & -\lambda & 6 \\ 0 & 0 & 2-\lambda \end{vmatrix} = (3-\lambda) \cdot (-1)^{1+1} \cdot \begin{vmatrix} -\lambda & 6 \\ 0 & 2-\lambda \end{vmatrix}$$

$$= \boxed{(3-\lambda) \cdot (-\lambda) \cdot (2-\lambda) = P_A(\lambda)}$$

✓ characteristic polynomial.

$$(A - \lambda \mathbb{I}_n) \vec{x} = \vec{0}$$

✓ 1. characteristic polynomial,

2. eigenvalues,

3. eigenvectors.

$$\Rightarrow P_A(\lambda) = (3-\lambda)(2-\lambda) \cdot (-\lambda) = 0$$

$$\Rightarrow \boxed{\lambda_1 = 0, \lambda_2 = 2, \lambda_3 = 3}$$

✓ eigenvalues.

$$\underline{\lambda_1 = 0}: (A - 0 \cdot \mathbb{I}_3) \cdot \vec{x} = \vec{0} \Rightarrow A \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3x_1 + 6x_2 - 8x_3 \\ 6x_3 \\ 2x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow x_3 = 0 \Rightarrow 3x_1 + 6x_2 = 0 \Rightarrow x_1 = -2x_2$$

$$\left\{ (-2x_2, x_2, 0) : x_2 \in \mathbb{R} \right\} \text{ eigenvectors corresponding } \lambda_1 = 0.$$

Finding Eigenvalues and Eigenvectors

Solution (1)

$$P_A(\lambda) = \det(A - \lambda I_n)$$

$$A - \lambda I_3 = \begin{pmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\underline{\lambda_2 = 2}; \quad (A - 2I_3) \cdot \vec{x} = \vec{0} \Rightarrow \begin{pmatrix} 1 & 6 & -8 \\ 0 & -2 & 6 \\ 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 + 6x_2 - 8x_3 \\ -2x_2 + 6x_3 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$-2x_2 + 6x_3 = 0 \Rightarrow x_2 = 3x_3 \Rightarrow x_1 + 6 \cdot 3x_3 - 8x_3 = x_1 + 10x_3 = 0 \\ \Rightarrow x_1 = -10x_3$$

$$(-10x_3, 3x_3, x_3) = x_3 \cdot (-10, 3, 1)$$

$\{ +(-10, 3, 1) : t \in \mathbb{R} \}$ eigenvectors corresponding to $\lambda = 2$.

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$$\underline{\lambda_3 = 3:}$$

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$$(A - \lambda I_3) \vec{x} = \vec{0}$$

$$\begin{pmatrix} 0 & 6 & -8 \\ 0 & -3 & 6 \\ 0 & 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 - \lambda & 6 & -8 \\ 0 & -\lambda & 6 \\ 0 & 0 & 2 - \lambda \end{pmatrix}$$

$$\Rightarrow -x_3 = 0$$

$$\Rightarrow \begin{cases} -x_3 = 0 \\ -3x_2 + 6x_3 = -3x_2 = 0 \Rightarrow x_2 = 0 \end{cases}$$

$$(x_1, 0, 0) = x_1 \cdot (1, 0, 0)$$

$$\left\{ \pm (1, 0, 0) : t \in \mathbb{R} \right\} \text{ eigenvectors corresponding } \lambda_3 = 3.$$

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Finding Eigenvalues and Eigenvectors

Solution (2)

Since, eigenvalues are the roots of $P_A(\lambda)$, we have

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Since, eigenvalues are the roots of $P_A(\lambda)$, we have

$$\begin{aligned}P_A(\lambda) &= -\lambda(3-\lambda)(2-\lambda) = 0 \\ \Rightarrow \lambda_1 &= 0, \lambda_2 = 2, \lambda_3 = 3.\end{aligned}$$

Finding Eigenvalues and Eigenvectors

Solution (3)

For $\lambda_1 = 0$,

$$(A - 0I_3) = A = \begin{pmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

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$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = t \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, t \in \mathbb{R} \Rightarrow \vec{v}_1 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}.$$

Finding Eigenvalues and Eigenvectors

Solution (3)

For $\lambda_2 = 2$,

$$(A - 2I_3) = \begin{pmatrix} 1 & 6 & -8 \\ 0 & -2 & 6 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 10 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{pmatrix}$$

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Solution (3)

For $\lambda_3 = 3$,

$$(A - 3I_3) = A - \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 6 & -8 \\ 0 & -3 & 6 \\ 0 & 0 & -1 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

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Cayley-Hamilton Theorem

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Theorem (Cayley-Hamilton Theorem)

Every matrix A is a root of its characteristic polynomial, i.e.,

$$P_A(A) = O_{n \times n}.$$

$$P_A(\lambda) = \det(A - \lambda I_n).$$

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$$P_A(A) = A^n + a_{n-1}A^{n-1} + \dots + a_1A + a_0I_n = 0$$

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$$P_A(A) = A^n + a_{n-1}A^{n-1} + \dots + a_1A + a_0I_n = 0$$

$$\Rightarrow A^n = -(a_{n-1}A^{n-1} + \dots + a_1A + a_0I_n)$$

$$A^n + a_{n-1}A^{n-1} + \dots + a_1A + a_0I_n = 0, \quad a_0 \neq 0 \quad (|A| \neq 0)$$

$$\frac{a_0 \cdot I_n}{a_0} = - \frac{(A^n + a_{n-1}A^{n-1} + \dots + a_1A)}{a_0} = A \cdot \frac{-(A^{n-1} + a_{n-1}A^{n-2} + \dots + a_1I_n)}{a_0}$$

$$I_n = A \cdot \left[\frac{-1}{a_0} (A^{n-1} + a_{n-1}A^{n-2} + \dots + a_1I_n) \right] \Rightarrow \underline{\underline{A^{-1}}}$$

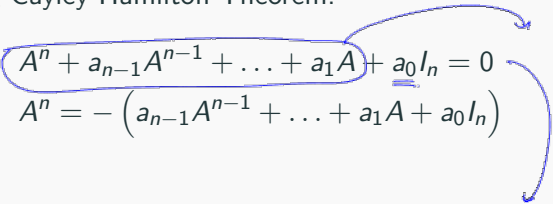
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Note: The inverse or any power of a square matrix can be computed by using Cayley-Hamilton Theorem.

$$\begin{aligned} P_A(A) &= A^n + a_{n-1}A^{n-1} + \dots + a_1A + \underline{a_0I_n} = 0 \\ \Rightarrow A^n &= -\left(a_{n-1}A^{n-1} + \dots + a_1A + a_0I_n\right) \end{aligned}$$


If $a_0 \neq 0$, then

$$I_n = A \underbrace{\frac{-1}{a_0} \left(A^{n-1} + a_{n-1}A^{n-2} + \dots + a_1I_n \right)}_{A^{-1}}$$

Example

$$A - \lambda \cdot \mathbb{I}_2 = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{pmatrix} \Rightarrow p_A(\lambda) = \begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix}$$

$$= (1-\lambda)(3-\lambda) - 8$$

$$= 3 + \lambda^2 - 4\lambda - 8 = \lambda^2 - 4\lambda - 5$$

$$\Rightarrow \boxed{p_A(\lambda) = \lambda^2 - 4\lambda - 5} \text{ charakteristisches Polynom.}$$

Example

Let $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$. Find

$$(A - \lambda \mathbb{I}_n) \cdot \vec{x} = \vec{0}$$

1. characteristic polynomial,

$$\Rightarrow p_A(\lambda) = \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1) = 0$$

$$\Rightarrow \lambda_1 = 5, \lambda_2 = -1 \text{ (eigenvalues)}$$

$$\underline{\lambda_1 = 5}: (A - 5 \cdot \mathbb{I}_2) \cdot \vec{x} = \vec{0}$$

$$\begin{pmatrix} -4 & 4 \\ 2 & -2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow -4x_1 + 4x_2 = 0 \rightarrow x_1 = x_2$$

$$(x_1, x_1) = x_1 \cdot (1, 1)$$

$$\underline{\left\{ t \cdot (1, 1) : t \in \mathbb{R} \right\} \text{ eigenvectors for } \lambda = 5}$$

$$\underline{\lambda_2 = -1}: (A + 1 \cdot \mathbb{I}_2) \cdot \vec{x} = \vec{0}$$

$$\begin{pmatrix} 2 & 4 \\ 2 & 4 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow 2x_1 + 4x_2 = 0 \rightarrow x_1 = -2x_2$$

$$(-2x_2, x_2) = x_2 \cdot (-2, 1)$$

$$\underline{\left\{ t \cdot (-2, 1) : t \in \mathbb{R} \right\} \text{ eigenvectors for } \lambda = -1.}$$

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Let $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$. Find

1. characteristic polynomial,
2. eigenvalues,
3. eigenvectors,
4. matrices P and D such that $A = PDP^{-1}$, if any,

Example

$$5) \quad p_A(\lambda) = \lambda^2 - 4\lambda - 5 \Rightarrow p_A(A) = A^2 - 4A - 5I_2 = O_{2 \times 2}$$

$$A^2 - 4A = 5 \cdot I_2 \Rightarrow A \cdot (A - 4 \cdot I_2) = 5 \cdot I_2$$

Example

Let $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$. Find

$$A \cdot \frac{(A - 4I_2)}{5} = A^{-1} \Rightarrow A^{-1} = \frac{1}{5} \cdot (A - 4I_2)$$

$$A^{-1} = \frac{1}{5} \cdot \begin{pmatrix} -3 & 4 \\ 2 & -1 \end{pmatrix}$$

1. characteristic polynomial,
2. eigenvalues,
3. eigenvectors,
- ~~4. matrices P and D such that $A = PDP^{-1}$, if any,~~
5. A^{-1} and A^5 (by using Cayley-Hamilton Theorem).

$$A^2 - 4A - 5I_2 = 0 \Rightarrow A^2 = 4A + 5I_2 \Rightarrow A^4 = A^2 A^2 = (4A + 5I_2)(4A + 5I_2)$$

$$= (16A^2 + 40A + 25I_2) = 16(4A + 5I_2) + 40A + 25I_2 = 104A + 105I_2$$

$$= A^5 = A A^4 = A(104A + 105I_2) = 104A^2 + 105A = 104(4A + 5I_2) + 105A$$

$$A^5 = 521A + 525I_2$$

Finding Eigenvalues and Eigenvectors

Solution (1)

$$P_A(\lambda) = \det(A - \lambda I_n)$$

$$A - \lambda I_2 = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Finding Eigenvalues and Eigenvectors

Solution (1)

$$P_A(\lambda) = \det(A - \lambda I_n)$$

$$\begin{aligned} A - \lambda I_2 &= \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 - \lambda & 4 \\ 2 & 3 - \lambda \end{pmatrix} \end{aligned}$$

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$$\begin{aligned} A - \lambda I_2 &= \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 - \lambda & 4 \\ 2 & 3 - \lambda \end{pmatrix} \\ \Rightarrow P_A(\lambda) &= (1 - \lambda)(3 - \lambda) - 8 \\ &= \lambda^2 - 4\lambda - 5. \end{aligned}$$

Finding Eigenvalues and Eigenvectors

Solution (2)

The eigenvalues are the roots of $P_A(\lambda)$ such that

$$P_A(\lambda) = \lambda^2 - 4\lambda - 5 = 0$$

Finding Eigenvalues and Eigenvectors

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The eigenvalues are the roots of $P_A(\lambda)$ such that

$$\begin{aligned}P_A(\lambda) &= \lambda^2 - 4\lambda - 5 = 0 \\ \Rightarrow \lambda_1 &= -1, \lambda_2 = 5.\end{aligned}$$

Finding Eigenvalues and Eigenvectors

Solution (3)

For $\lambda_1 = -1$,

$$(A + I_2) = \begin{pmatrix} 2 & 4 \\ 2 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$$

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$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = t \begin{pmatrix} -2 \\ 1 \end{pmatrix}, t \in \mathbb{R} \Rightarrow \vec{v}_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}.$$

Finding Eigenvalues and Eigenvectors

Solution (3)

For $\lambda_2 = 5$,

$$(A - 5I_2) = \begin{pmatrix} -4 & 4 \\ 2 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$$

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Finding Eigenvalues and Eigenvectors

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$$\vec{x} = \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow (A - 5I_2) \vec{x} = \vec{0} \Rightarrow x - y = 0$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = t \begin{pmatrix} 1 \\ 1 \end{pmatrix}, t \in \mathbb{R} \Rightarrow \vec{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Finding Eigenvalues and Eigenvectors

Solution (4)

$$P = \begin{pmatrix} -2 & 1 \\ 1 & 1 \end{pmatrix}, P^{-1} = \begin{pmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix} \text{ and } D = \begin{pmatrix} -1 & 0 \\ 0 & 5 \end{pmatrix} \Rightarrow$$

$$A = PDP^{-1}.$$

Cayley-Hamilton Theorem

Solution (5)

$$P_A(A) = A^2 - 4A - 5I_2 = 0$$

Cayley-Hamilton Theorem

Solution (5)

$$\begin{aligned}P_A(A) &= A^2 - 4A - 5I_2 = 0 \\ \Rightarrow \underbrace{A \frac{1}{5} (A - 4I_2)}_{A^{-1}} &= I_2\end{aligned}$$

Cayley-Hamilton Theorem

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$$P_A(A) = A^2 - 4A - 5I_2 = 0$$

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$$A^{-1} = \frac{1}{5} \left(\begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \right) = \frac{1}{5} \begin{pmatrix} -3 & 4 \\ 2 & -1 \end{pmatrix}$$

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$$A^2 = 4A + 5I_2$$

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$$A^4 = (4A + 5I_2)(4A + 5I_2)$$

Cayley-Hamilton Theorem

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$$A^5 = A(4A + 5I_2)(4A + 5I_2)$$

?